Categories, Proofs, and Processes: Assignment 3

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A functor $F : \mathcal{C} \to \mathcal{D}$ is

- **faithful** if each map $F_{A,B} : \mathcal{C}(A, B) \to \mathcal{D}(F\ A, F\ B)$ is injective;
- **full** if each map $F_{A,B} : \mathcal{C}(A, B) \to \mathcal{D}(F\ A, F\ B)$ is surjective

Question 1

For each function $f : C \to D$ in **Set**, S maps it to the functor

$$S\ f : \mathcal{C} \to \mathcal{D}$$

where $\mathcal{C}, \mathcal{D}$ are discrete categories in **Cat** with $|C|, |D|$ objects respectively.

$$\mathbf{Set} \xrightarrow{S} \mathbf{Cat}$$

$$\begin{tikzcd} C \arrow[r, mapsto] \arrow[d, "f"'] & \mathcal{C} \arrow[d, "S\ f"] \\ {} \arrow[r, mapsto, shorten=1em] & {} \\ D \arrow[r, mapsto] & \mathcal{D} \end{tikzcd}$$

Then $\forall a \in C, b \in D, (S\ a) \in \mathrm{Ob}(\mathcal{C}), (S\ b) \in \mathrm{Ob}(\mathcal{D})$, the mapping is

$$S\ f :: S\ a \mapsto S\ b \iff f :: a \mapsto b$$

$$S\ f :: 1_{(S\ a)} \mapsto 1_{(S\ b)} \iff f :: a \mapsto b$$

Proposition 1.1. *$\forall f \in Ar(\boldsymbol{Set}), (S\ f) : \mathcal{C} \to \mathcal{D}$ satisfies functoriality.*

Proof. Since there is a valid function where $\forall a \in C, f(a) = b$, $(S\ f)$ will map every object $(S\ a)$ in $\mathcal{C}$ to an $(S\ b)$ in $\mathcal{D}$.

There are no arrows between objects in the discrete categories, only identity arrows. For any identity arrow 1_A in $\mathcal{C}$,

$$(S\ f)(1_A \circ 1_A) = (S\ f)(1_A) = 1_{(S\ f)(1_A)} = (S\ f)(1_A) \circ (S\ f)(1_A)$$

$\square$

Fact 1.2. The functor $S : \mathbf{Set} \to \mathbf{Cat}$ satisfies functoriality.

Proof. Every object set A in **Set** will get mapped to a discrete category $S\ A$ in **Cat**. For every pairs of functions $f : A \to B, g : B \to C$ in **Set**, and $a \in A, c \in C$,

$$\begin{aligned}
(S(g \circ f)) :: S\ a \mapsto S\ c &\iff g \circ f(a) = c \\
&\iff \exists b \in B : f(a) = b \wedge g(b) = c \\
&\iff \exists b \in B : S\ f :: S\ a \mapsto S\ b \wedge S\ g :: S\ b \mapsto S\ c \\
&\iff (S\ g \circ S\ f) :: S\ a \mapsto S\ c
\end{aligned} \tag{1}$$

$$S\ 1_A :: S\ a \mapsto S\ a = 1_{S\ A}$$

Then functoriality is satisfied. □

Proposition 1.3. *The functor $S : \boldsymbol{Set} \to \boldsymbol{Cat}$ is full and faithful.*

Proof. To show faithfulness, let $f, g : C \to D$ be arbitrary functions in **Set** so $S\ f, S\ g : \mathcal{C} \to \mathcal{D}$. Suppose $S\ f = S\ g$. Then since for all $a \in C$,

$$S\ f :: S\ a \mapsto S\ b \iff f :: a \mapsto b$$

$$S\ g :: S\ a \mapsto S\ b \iff g :: a \mapsto b$$

we have

$$\forall a \in C : f :: a \mapsto b \iff S\ f = S\ g :: S\ a \mapsto S\ b \iff g :: a \mapsto b$$

and $f = g$. S is faithful.

For fullness, let $F : \mathcal{C} \to \mathcal{D}$ be an arbitrary functor in $\mathbf{Cat}(S\ A, S\ B)$, where A, B are sets in **Set**. Let $f : A \to B$. For each object $S\ a$ in $S\ A$ where $F :: S\ a \mapsto S\ b$, we can have some $a \in A, f(a) = b$ since $S\ A$ has $|A|$ objects. Then $S\ f = F$. So S is full.

□

Question 2

Definition 2.4. The bivariant Hom-functor is given by

$$\mathcal{C}(\_,\_)(A,B) := \mathcal{C}(A,B)$$

$$\mathcal{C}(\_,\_)(g : B \to A, f : C \to D) := h \mapsto f \circ h \circ g, \forall h \in \mathcal{C}(A,C)$$

for all objects and arrows $A, B, C, D \in \mathrm{Ob}(\mathcal{C}), f, g \in \mathrm{Ar}(\mathcal{C})$.

Proposition 2.5. *The bivariant Hom-functor is a functor.*

Proof. Let $A, B, C, D, E, F \in \mathrm{Ob}(\mathcal{C}) = \mathrm{Ob}(\mathcal{C}^{\mathrm{op}})$ be arbitrary objects.

For any object pair A, B, the functor has a mapping to the set $\mathcal{C}(A,B)$.

Let $f : A \to B, g : B \to C$ be arbitrary arrows in $\mathcal{C}$, and $h : E \to D, k : F \to E$ be arbitrary arrows in $\mathcal{C}^{\mathrm{op}}$.

$$\begin{aligned}\mathcal{C}(\_,\_)(h,g) \circ \mathcal{C}(\_,\_)(k,f) &= (p \mapsto g \circ p \circ h) \circ (p' \mapsto f \circ p' \circ k) \\ &= (p \mapsto g \circ (f \circ p \circ k) \circ h) \\ &= (p \mapsto (g \circ f) \circ p \circ (k \circ h)) \\ &= \mathcal{C}(\_,\_)(h \circ k, g \circ f) \\ &= \mathcal{C}(\_,\_)((h,g) \circ (k,f))\end{aligned} \tag{2}$$

So the composition law is satisfied.

For the identity laws,

$$\begin{aligned}\mathcal{C}(\_,\_)(1_A, 1_B) &= \mathcal{C}(1_A, 1_B) \\ &= h \mapsto 1_B \circ h \circ 1_A \\ &= h \mapsto h \\ &= 1_{\mathcal{C}(A,B)} \\ &= 1_{\mathcal{C}(\_,\_)(A,B)}\end{aligned} \tag{3}$$

Thus, the bivariant Hom-functor satisfies *functoriality*.

□

Question 3

(a)

Fact 3.6. Faithful functors do not, in general, reflect isomorphisms.

Example 3.7.

$$A \xrightarrow{f} B \qquad\qquad F\ A \underset{g=(F\ f)^{-1}}{\overset{F\ f}{\rightleftarrows}} F\ B$$

The diagram above shows two-object categories $\mathcal{C}, \mathcal{D}$ respectively, with the identities assumed but not drawn. The arrow $g = (F\ f)^{-1} : F\ B \to F\ A$ in $\mathcal{D}$ has no mapping from an arrow in $\mathcal{C}$.

Functoriality is satisfied since

$$F(1_A) = 1_{F\ A}, \quad F(1_B) = 1_{F\ B}$$

$$F(f \circ 1_A) = F(f) = F(f) \circ 1_{F\ A} = F(f) \circ F(1_A)$$
$$F(1_B \circ f) = F(f) = 1_{F\ B} \circ F(f) = F(1_B) \circ F(F)$$

The functor $F : \mathcal{C} \to \mathcal{D}$ is faithful since

$$F\ f \neq F\ 1_A = 1_{F\ A} \neq F\ 1_B = 1_{F\ B}$$

It can be noted it is not full because of the arrow g in $\mathcal{D}$.

The arrow $F\ f$ is isomorphic (to $(F\ f)^{-1}$) but f is not iso. The functor does not reflect isomorphism.

(b)

Proposition 3.8. *Every full and faithful functor reflects isomorphisms.*

Proof. Let the functor $F : \mathcal{C} \to \mathcal{D}$ be full and faithful. By fullness, for any arrow g in $\mathcal{D}$, there exists an arrow f in $\mathcal{C}$ such that $F\ f = g$.

Suppose we have two iso arrows g, g^{-1} in $\mathcal{C}$. Using fullness we find $f : A \to B, f' : B' \to A'$ such that $F\ f = g, F\ f' = g^{-1}$. Since g, g^{-1} are iso, $F\ A = F\ A', F\ B = F\ B'$ so

$$g : F\ A \to F\ B, \quad g^{-1} : F\ B \to F\ A$$

where

$$g^{-1} \circ g = 1_{F\ A}, \quad g \circ g^{-1} = 1_{F\ B}$$

By functoriality, we have from composition that

$$F(f \circ f') = F\ f \circ F\ f' = g \circ g^{-1} = 1_{F\ B}$$

$$F(f' \circ f) = F\ f' \circ F\ f = g^{-1} \circ g = 1_{F\ A}$$

We also have from the identity law that

$$F(1_A) = 1_{F\ A}, \quad F(1_B) = 1_{F\ B}$$

By faithfulness,

$$F(1_A) = F(f' \circ f) = 1_{F\ A} \implies f' \circ f = 1_A$$
$$F(1_B) = F(f \circ f') = 1_{F\ B} \implies f \circ f' = 1_B$$

So $f' = f^{-1}$, and f, f^{-1} are iso.

Given an iso arrow $F\ f$ in $\mathcal{D}$, we then have an iso arrow f in $\mathcal{C}$. The full and faithful functor reflects isomorphisms.

□

(c)

Proposition 3.9. *Every equivalence preserves monics.*

Proof. Let $F : \mathcal{C} \to \mathcal{D}$ be an equivalence, and let $f : B \to C$ be a monic arrow in $\mathcal{C}$. Then $F\ f : F\ B \to F\ C$. Using fullness, let $F\ g, F\ h : A' \to F\ B$ be arbitrary arrows in $\mathcal{D}$.

By equivalence, there exists an object A of $\mathcal{C}$ such that $F(A) \cong A'$.

By fullness, let the arrows $g, h : A \to B$ be in $\mathcal{C}$.

Suppose $F\ f \circ F\ g = F\ f \circ F\ h$. Then

$$\begin{aligned} F\ f \circ F\ g = F\ f \circ F\ h &\implies F(f \circ g) = F(f \circ h) && \text{(by functoriality)} \\ &\implies f \circ g = f \circ h && \text{(by faithfulness)} \\ &\implies g = h && \text{(since } f \text{ is monic)} \\ &\implies F\ g = F\ h \end{aligned} \tag{4}$$

Then

$$\forall F\ g, F\ h : F\ f \circ F\ g = F\ f \circ F\ h \implies F\ g = F\ h$$

so $F\ f$ is monic and the equivalence preserves monics. □

Question 4

Proposition 4.10. *If a category $\mathcal{C}$ has a terminal object and pullbacks, it has binary products and equalisers.*

Proof. Let T be the terminal object, and let $\tau_A : A \to T$ denote the unique arrow from an object A to T.

$$\begin{tikzcd}
D' \arrow[drr, bend left, "q'"] \arrow[dr, dashed, "h'"] \arrow[ddr, bend right, "p'"'] & & \\
& D \arrow[r, "q"] \arrow[d, "p"'] & B \arrow[d, "\tau_B"] \\
& A \arrow[r, "\tau_A"'] & T
\end{tikzcd}$$

The morphisms $A \xrightarrow{\tau_A} T \xleftarrow{\tau_B} B$ then have the pullback $A \xleftarrow{p} D \xrightarrow{q} B$.

For any morphisms $A \xleftarrow{p'} D' \xrightarrow{q'} B$, we have by composition,

$$\tau_A \circ p' = \tau_{D'} = \tau_B \circ q' \implies h' : D' \to D \text{ is unique}$$

such that $p' = p \circ h'$ and $q' = q \circ h'$. Then D with morphisms $A \xleftarrow{p} D \xrightarrow{q} B$ is the product of A and B, since for all pairs $A \xleftarrow{p'} D' \xrightarrow{q'} B$, there is the unique morphism $h' = \langle p', q' \rangle : D' \to D$ such that $p' = p \circ h'$ and $q' = q \circ h'$.

Using the binary product, we can then show equalisers. Suppose we have the parallel morphisms $A \underset{g}{\overset{f}{\rightrightarrows}} B$. As shown before, the product $A \times B$ with pullback morphisms $A \xleftarrow{\pi_1} A \times B \xrightarrow{\pi_2} B$ exist. The following diagram with f commutes, and is similar if g is used instead of f, where 1_A is the identity on A.

$$\begin{tikzcd}
A & A \times B \arrow[l, "\pi_1"'] \arrow[r, "\pi_2"] & B \\
& A \arrow[ul, "1_A"] \arrow[u, dashed, "\langle 1_A, f \rangle"'] \arrow[ur, "f"'] &
\end{tikzcd}$$

Let $A \xleftarrow{p} E \xrightarrow{q} A$ be the pullback of $A \xrightarrow{\langle 1_A, f \rangle} A \times B \xleftarrow{\langle 1_A, g \rangle} A$. So $\langle 1_A, f \rangle \circ p = \langle 1_A, g \rangle \circ q$.

$$\begin{tikzcd}
E \arrow[r, "q"] \arrow[d, "p"'] & A \arrow[d, "\langle 1_A, g \rangle"] \\
A \arrow[r, "\langle 1_A, f \rangle"'] & A \times B
\end{tikzcd}$$

Then we show that $e = p = q : E \to A$ is the equaliser of (f, g). First, we show $p = q$ using the product,

$$\begin{aligned} p &= \pi_1 \circ \langle 1_A, f \rangle \circ p \\ &= \pi_1 \circ \langle 1_A, g \rangle \circ q \\ &= q \end{aligned} \tag{5}$$

Then we have

$$
\begin{aligned}
f \circ e &= \pi_2 \circ \langle 1_A, f\rangle \circ e \\
&= \pi_2 \circ \langle 1_A, g\rangle \circ e \qquad (6)\\
&= g \circ e
\end{aligned}
$$

Suppose we then have a morphism $h : C \to A$ such that $f \circ h = g \circ h$. Then

$$
\begin{aligned}
\langle 1_A, f\rangle \circ h &= \langle 1_A \circ h, f \circ h\rangle \\
&= \langle 1_A \circ h, g \circ h\rangle \qquad (7)\\
&= \langle 1_A, g\rangle \circ h
\end{aligned}
$$

Then by the pullback, there is a unique morphism $\hat{h} : C \to E$ such that $h = e \circ \hat{h}$.

$$
\begin{tikzcd}
E \arrow[r, "e"] & A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \\
C \arrow[u, dashed, "\hat{h}"] \arrow[ur, "h"'] & &
\end{tikzcd}
$$

Then $e : E \to A$, the pullback of $(\langle 1_A, f\rangle, \langle 1_A, g\rangle)$, is the equaliser of (f, g) where $A \overset{f}{\underset{g}{\rightrightarrows}} B$.

Thus, it is shown that the category $\mathcal{C}$ has binary products and equalisers.

∎

Question 5

Proposition 5.11. *If $ABCD, BEDF$ are pullback squares, then so is $AECF$.*

$$
\begin{tikzcd}
G \arrow[dr, dashed, "h"] \arrow[drr, dashed, "h'"] \arrow[drrr, bend left, "q"] \arrow[ddr, bend right, "p"'] & & & \\
& A \arrow[r, "a"] \arrow[d, "c"] & B \arrow[r, "b"] \arrow[d, "d"] & E \arrow[d, "e"] \\
& C \arrow[r, "f"'] & D \arrow[r, "g"'] & F
\end{tikzcd}
$$

Proof. Suppose we have morphisms $C \xleftarrow{p} G \xrightarrow{q} E$ such that $(g \circ f) \circ p = e \circ q$.

By the pullback square $BEDF$,

$$e \circ q = g \circ (f \circ p) \implies \exists! h' : G \to B, q = b \circ h' \wedge f \circ p = d \circ h'$$

By the pullback square $ABCD$ and using h',

$$d \circ h' = f \circ p \implies \exists! h : G \to A, p = c \circ h \wedge h' = a \circ h$$

From the unique h, h' such that $q = b \circ h', h' = a \circ h$, then h is the unique morphism where $q = b \circ a \circ h$. Then combining $q = (b \circ a) \circ h$ with $p = c \circ h$ (from pullback square $ABCD$) where h is unique, $AECF$ is also a pullback square. ∎

Question 6

The diagram D maps as follows:

$$D :: n \mapsto ([n], \leq), \quad (f : n \to m) \mapsto (g : ([n], \leq) \hookrightarrow ([m], \leq))$$

where g is the inclusion map from $[n]$ to $[m]$.

$$\begin{tikzcd} \mathcal{J} \arrow[r, "D"] & \textbf{Natural} \end{tikzcd}$$

$$\begin{tikzcd} n \arrow[r, mapsto] \arrow[d, "f"'] & ([n], \leq) \arrow[d, hook, "g"] \\ {} \arrow[r, mapsto] & {} \\ m \arrow[r, mapsto] & ([m], \leq) \end{tikzcd}$$

Each cone to D is an object $([n], \leq)$ in **Natural** and the family of arrows γ,

$$\{\gamma_I : ([n], \leq) \to (D\ I)\}_{I \in \mathrm{Ob}(\mathcal{J})} = \{\gamma_m : ([n], \leq) \hookrightarrow ([m], \leq)\}_{m \in \mathbb{N}, m \geq n}$$

Let $([p], \leq)$ with its family of arrows δ be an arbitrary cone to D. Since inclusion maps from $[p]$ to $[q]$ are only defined when $p \leq q$, consider $f : n \to m$ in $\mathcal{J}$, where $p \leq n \leq m$. We can see that

$$\forall i \in [p] : (D\ f) \circ \delta_n(i) = (D\ f)(i) = \delta_m(i) = i$$

so $(D\ f) \circ \delta_n = \delta_m$.

$$\begin{tikzcd} ([n], \leq) \arrow[rr, "D\ f"] & & ([m], \leq) \\ & ([p], \leq) \arrow[ul, "\delta_n"] \arrow[ur, "\delta_m"'] & \end{tikzcd}$$

The limit for D is the object $(\mathbb{N}, \leq)$ with its family of arrows containing only the identity, $\{1_\mathbb{N}\}$. For any other posets $([n], \leq), n \in \mathbb{N}$ in **Natural**, $[n] \subset \mathbb{N}$, so there is no mapping from $(\mathbb{N}, \leq)$ other than to itself.

$$\begin{tikzcd} & (\mathbb{N}, \leq) & \\ ([n], \leq) \arrow[ur, "\gamma_\mathbb{N}"] \arrow[rr, "\gamma_\mathbb{N}"'] & & (\mathbb{N}, \leq) \arrow[ul, "1_\mathbb{N}"'] \end{tikzcd}$$

The colimit for D is the object $([0], \leq)$ with its family of arrows

$$\{\gamma_n : ([0], \leq) \hookrightarrow ([n], \leq)\}_{n \in \mathbb{N}}$$

For any other posets $([n], \leq), n \in \mathbb{N}^+$ in **Natural**, $[0] \subset [n]$, so there is no mapping to $([0], \leq)$ other than from itself.